WAYGROUND formerly Quizizz **Worksheets**

Name

OOP - 7

Class

Total questions: 12

Date

Worksheet time: 7mins

1.

```
void f() {
    static int counter = 0;
    counter++;
    print("{} ", counter);
}

int main() {
    f();
    f();
    f();
}
```

a) 0 0 0

b) 1 2 3

c) 0 1 2

d) Няма да се компилира!

2.

```
void f() {
    int counter = 0;
    counter++;
    print("{} ", counter);
}

int main() {
    f();
    f();
    f();
}
```

a) 1 1 1

b) 1 2 3

c) 0 1 2

d) Няма да се компилира!

3.

```
class Human {
    static int idGenerator;
    int id;

public:
    Human() {
        idGenerator++;
        id = idGenerator;
    }
    int getIdGenerator() const {
        return idGenerator;
    }
};
int Human::idGenerator = 0;

int main() {
    Human h1;
    Human h2;
    print("{} ", h1.getIdGenerator());
    print("{}", h2.getIdGenerator());
}
```

a) 0 0

b) 1 2

c) 2 2

d) Няма да се компилира!

4.

```
class Human {
    static int objectCounter;
public:
    Human() {
        objectCounter++;
    }
    ~Human() {
        objectCounter--;
    }

    int getObjectCounter() {
        return objectCounter;
    }
};
int Human::objectCounter = 0;

int main() {
    Human* h1 = new Human();
    Human* h2 = new Human();
    print("{} ", Human::getObjectCounter());
    delete h1;
    print("{}", Human::getObjectCounter());
    delete h2;
}
```

a) 2 2

b) 2 1

c) 2 0

d) Няма да се компилира!

5.

```
class Human {
public:
    static int objectCounter;
    int age = 0;

    Human() {
        objectCounter++;
    }
    ~Human() {
        objectCounter--;
    }
    static int modifier() {
        objectCounter = 0;
        age = 10;
    }
};
int Human::objectCounter = 0;

int main() {
    Human h1;
    h1.modifier();
    print("{} {}", h1.objectCounter, h1.age);
}
```

a) 0 10

b) 0 0

c) 1 0

d) Няма да се компилира!

6.

```
class Human {
public:
    Human() {print("Human() ");}
    Human(const Human& other) {print("Human(other&) ");}
    Human(Human&& other) {print("Human(other&&) ");}
    ~Human() {print("~Human() ");}
};

int main() {
    Human h1;
    Human h2(h1);
}
```

a) Human() Human() ~Human() ~Human()

b) Human() Human(other&) ~Human() ~Human()

c) Human() Human(other&&) ~Human() ~Human()

d) Няма да се компилира!

7.

```
class Human {
public:
    Human() {print("Human() ");}
    Human(const Human& other) {print("Human(other&) ");}
    Human(Human&& other) {print("Human(other&&) ");}
    ~Human() {print("~Human() ");}
};

int main() {
    Human h1;
    Human h2(std::move(h1));
}
```

a) Human() Human(other&&) ~Human()

b) Human() Human(other&) ~Human() ~Human()

c) Human() Human(other&&) ~Human() ~Human()

d) Няма да се компилира!

8.

```
class Human {
public:
    Human() {print("Human() ");}
    Human(const Human& other) {print("Human(other&) ");}
    Human(Human&& other) {print("Human(other&&) ");}
    ~Human() {print("~Human() ");}
};

int main() {
    Human h1;
    Human h2(std::move(h1));
    Human h3(h1);
}
```

a) Human() Human(other&) ~Human() Human(other&&) ~Human() ~Human()

b) Human() Human(other&) Human(other&) ~Human() ~Human() ~Human()

c) Human() Human(other&&) Human(other&) ~Human() ~Human() ~Human()

d) Няма да се компилира!

9.

```
class Human {
public:
    Human() {print("Human() ");}
    ~Human() {print("~Human() ");}
};

int main() {
    Human arr[5];
}
```

a) 0 x Human()
0 x ~Human()

b) 5 x Human()
0 x ~Human()

c) 0 x Human()
5 x ~Human()

d) 5 x Human()
5 x ~Human()

e) Няма да се компилира!

10.

```
class Human {
public:
    Human() {print("Human() ");}
    ~Human() {print("~Human() ");}
};

int main() {
    Human* arr[5];
}
```

a) 0 x Human()
0 x ~Human()

b) 5 x Human()
0 x ~Human()

c) 0 x Human()
5 x ~Human()

d) 5 x Human()
5 x ~Human()

e) Няма да се компилира!

11.

```
class Human {
public:
    Human() {print("Human() ");}
    ~Human() {print("~Human() ");}
};

int main() {
    Human* arr = new Human[5];
}
```

a) 0 x Human()
0 x ~Human()

b) 5 x Human()
0 x ~Human()

c) 0 x Human()
5 x ~Human()

d) 5 x Human()
5 x ~Human()

e) Няма да се компилира!

12.

```
class Human {
public:
    Human() {print("Human() ");}
    ~Human() {print("~Human() ");}
};

int main() {
    Human** arr = new Human*[5];
}
```

a) 0 x Human()
0 x ~Human()

b) 5 x Human()
0 x ~Human()

c) 0 x Human()
5 x ~Human()

d) 5 x Human()
5 x ~Human()

e) Няма да се компилира!

Answer Keys

1. b) 1 2 3
2. a) 1 1 1
3. c) 2 2
4. d) Няма да се компилира!
5. d) Няма да се компилира!
6. b) Human() Human(other&)
   ~Human() ~Human()
7. c) Human()
   Human(other&&)
   ~Human() ~Human()
8. c) Human()
   Human(other&&)
   Human(other&)
   ~Human() ~Human()
   ~Human()
9. d) 5 x 5 x
   Human() ~Human()
10. a) 0 x 0 x
    Human() ~Human()
11. b) 5 x 0 x
    Human() ~Human()
12. a) 0 x 0 x
    Human() ~Human()